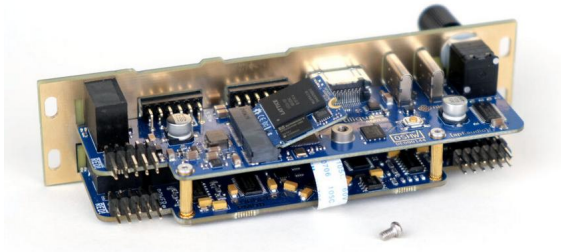


tiliqua: FPGA-based audio DSP platform

ORConf 2025

github.com/apfaudio/tiliqua

Seb Holzapfel / seb@apf.audio

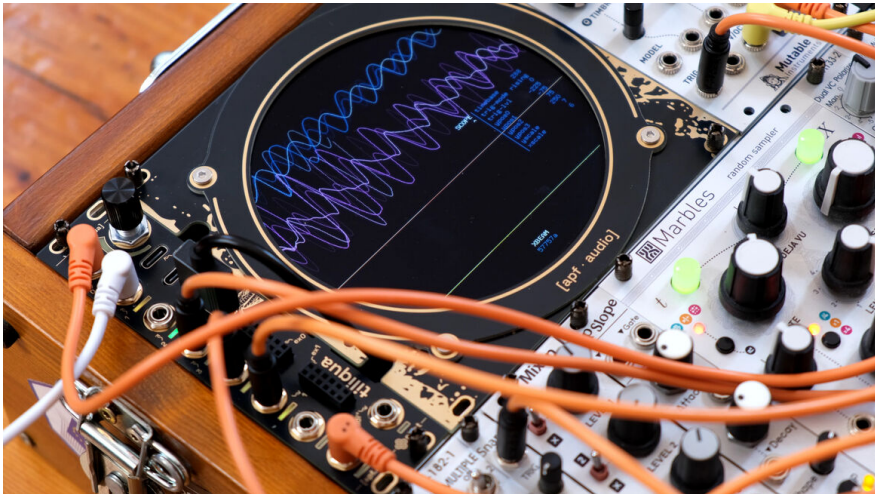


What is tiliqua?



- ▶ Open hardware platform for audio & video synthesis (Eurorack module)
- ▶ Open gateway library of DSP components (Amaranth HDL)
- ▶ Hot bitstream switching – easy to share designs.
- ▶ Useful to musicians, even without touching HDL.

Why Eurorack?



- ▶ Most popular format for modular synthesizers.
- ▶ **Ecosystem:** > 15,000 modules, > 1,000 different manufacturers.

4x IN
(192kHz, DC-coupled)

4x OUT
(192kHz, DC-coupled)



encoder

JTAG adapter

USB2 high-speed

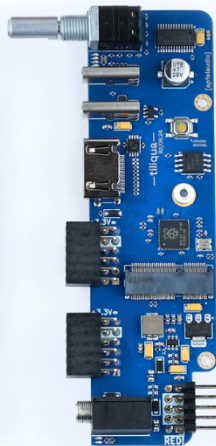
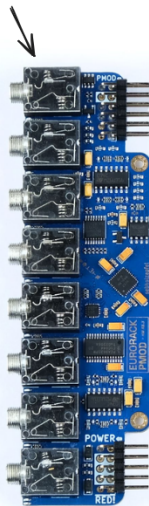
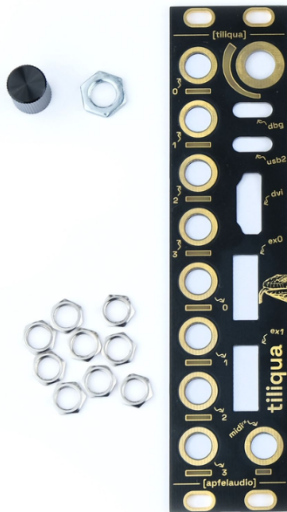
display output

2x expansion PMODs

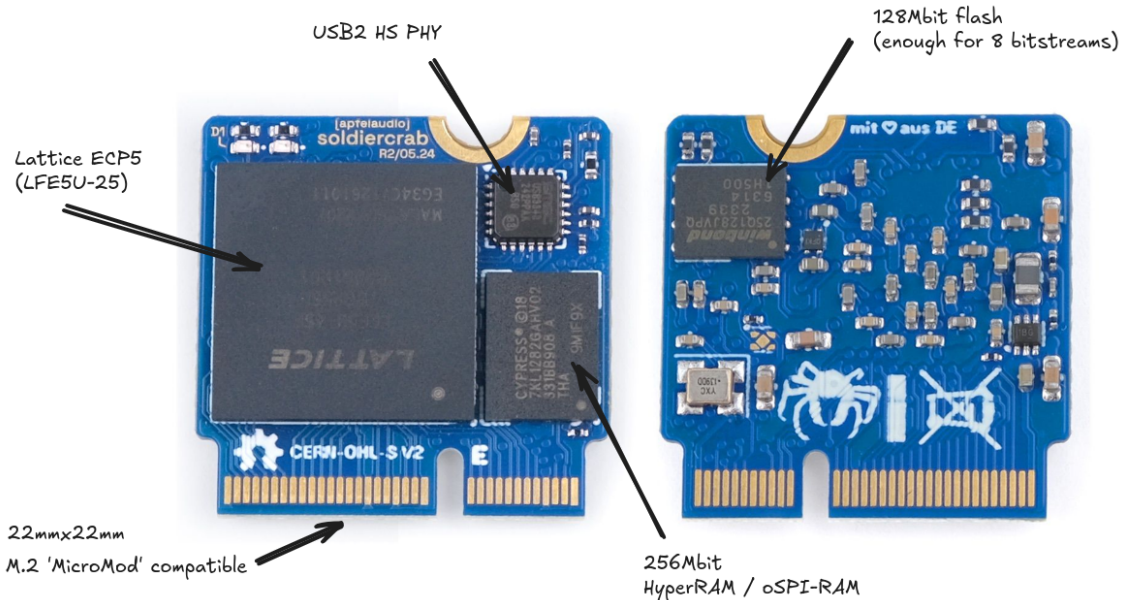
MIDI TRS IN

Audio Interface

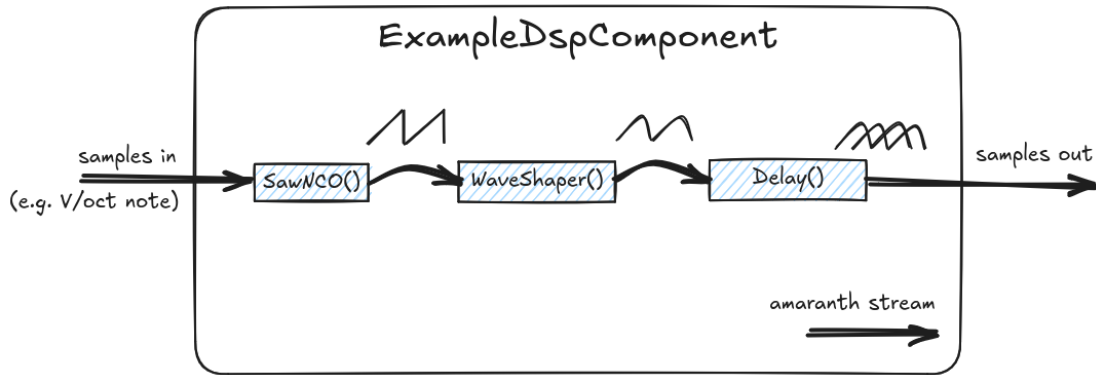
"SoldierCrab" ECP5 SoM



Tiliqua Motherboard



DSP pipelines, Amaranth HDL and streams



What's new?

- ▶ **More DSP cores** – frequency-domain stuff!
- ▶ **Streaming / visualization.**
- ▶ **USB2 Host** – in pure gateware.
- ▶ **Display hot-swapping** – dynamic clocking.
- ▶ **Tiny GPU** – font/line acceleration.
- ▶ **Auto-restore** – persist settings to config flash.

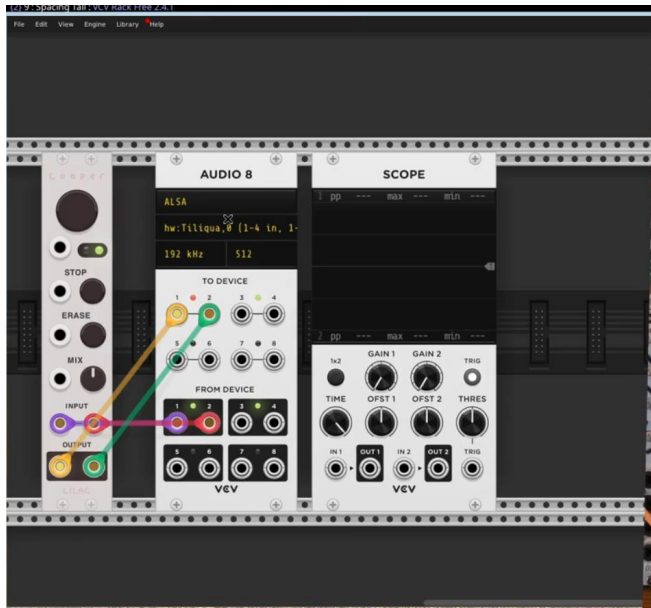
tiliqua DSP library - frequency-domain processing

- ▶ **FFT** – forward + inverse, iterative, low area usage.
- ▶ **CORDIC** – complex/polar conversions.
- ▶ **Window** – windowing core and selectable window functions.
- ▶ **STFT** – forward + inverse, frequency-domain processing of time-domain signals.
- ▶ **SpectralEnvelope** – compute filtered power spectra over time.
- ▶ **SpectralCrossSynthesis** – vocoder-like effect.

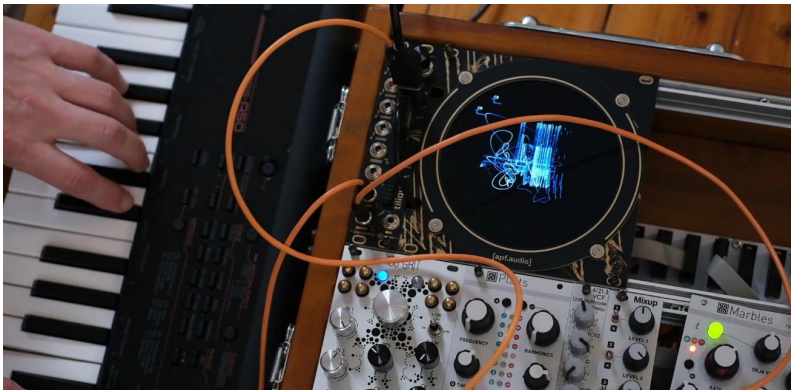
Also: check out Amaranth lib.fixed

- ▶ **Extension to Amaranth standard library**
- ▶ **RFC + Implementation (WIP):** github.com/amaranth-lang/amaranth/pull/1578
- ▶ **Comments welcome!**

Streaming / visualization



USB2 Host



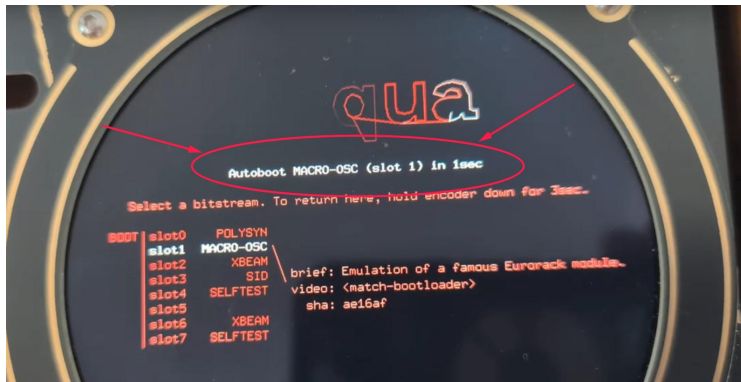
- ▶ Cheapest MIDI keyboards only have USB MIDI :(
- ▶ Solution: pure-gateway USB2 high-speed host. (first?)
- ▶ Details: github.com/apfaudio/tiliqua/pull/65
- ▶ Builds on LUNA: github.com/greatscottgadgets/luna

Display hot-swap / dynamic clocking



- ▶ People have different screens (pixel clock).
- ▶ Same bitstream should work everywhere.
- ▶ Core reads display EDID, dynamically adapts resolution!
- ▶ Details: github.com/apfaudio/tiliqua/pull/107

Auto-restore / persistent options



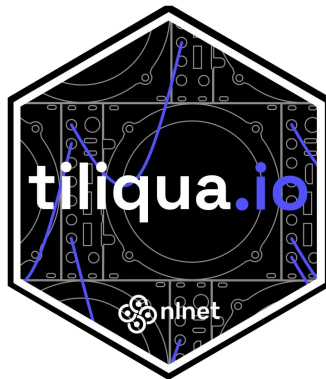
- ▶ Wear-levelled persistent storage to configuration flash
- ▶ Store last bitstream, all options, autoboot.
- ▶ Details: github.com/apfaudio/tiliqua/pull/129

Tiny embedded-graphics GPU (new!)



- ▶ Text, line, raster accelerator for embedded-graphics.
- ▶ Reduces bitstream size even more!.

Thank you... Funding!



Thank you... Preorders!

CROWD SUPPLY

BROWSE

APPLY

ABOUT

Search

Tiliqua

A powerful, hackable FPGA-based audio multitool for Eurorack



\$46,763 raised
of \$12,000 goal

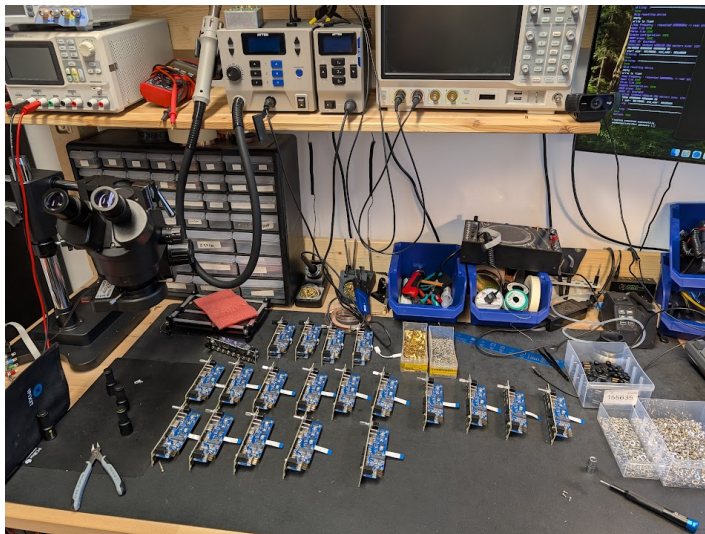
389% Funded!

6
updates

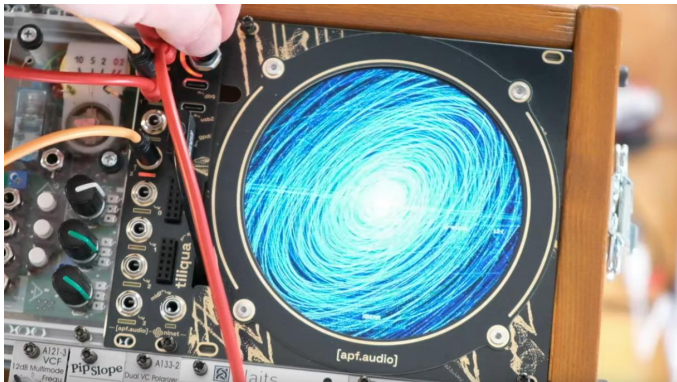
Jun 19
funded on

Available for pre-order.

Production!



Thank you for listening!



- ▶ contact: seb@apf.audio
- ▶ repository: github.com/apfaudio/tiliqua
- ▶ mailing list: apf.audio
- ▶ crowdsupply.com/apfaudio/tiliqua